

# XIAOMENG XU

<https://xxm19.github.io/> | xuxm@stanford.edu

## EDUCATION

---

**Stanford University**

09/2023-now

*PhD Candidate in Electrical Engineering, Advisor: Shuran Song*

**Stanford University**

09/2023-06/2025

*Masters in Electrical Engineering*

**Tsinghua University**

08/2019-06/2023

*Bachelor of Engineering in Automation (GPA: 3.96/4.0), Bachelor of Arts in Product Design*

## PUBLICATIONS

---

1. **Xiaomeng Xu**, Jisang Park, Han Zhang, Eric Cousineau, Aditya Bhat, Jose Barreiros, Dian Wang, Jeannette Bohg, Shuran Song, *HoMMI: Learning Whole-Body Mobile Manipulation from Human Demonstrations*. Robotics: Science and Systems (RSS 2026). [Paper]
2. Ryan Punamiya, Simar Kareer, Zeyi Liu, Josh Citron, Ri-Zhao Qiu, Xiongyi Cai, Alexey Gavryushin, Jiaqi Chen, Davide Liconti, Lawrence Y. Zhu, Patcharapong Aphiwetsa, Baoyu Li, Aniketh Cheluva, Pranav Kuppili, Yangcen Liu, Dhruv Patel, Aidan Gao, Hye-Young Chung, Ryan Co, Renee Zbizika, Jeff Liu, **Xiaomeng Xu**, Haoyu Xiong, Geng Chen, Sebastiano Olliani, Chenyu Yang, Xi Wang, James Fort, Richard Newcombe, Josh Gao, Jason Chong, Garrett Matsuda, Aseem Doriwala, Marc Pollefeys, Robert Katzschmann, Xiaolong Wang, Shuran Song, Judy Hoffman, Danfei Xu, *EgoVerse: An Egocentric Human Dataset for Robot Learning from Around the World*. Robotics: Science and Systems (RSS 2026). [Paper]
3. Hojung Choi\*, Yifan Hou\*, Chuer Pan, Seongheon Hong, Austin Patel, **Xiaomeng Xu**, Mark Cutkosky, Shuran Song, *In-the-Wild Compliant Manipulation with UMI-FT*. IEEE International Conference on Robotics and Automation (ICRA 2026). [Paper]
4. **Xiaomeng Xu\***, Yifan Hou\*, Chendong Xin, Zeyi Liu, Shuran Song, *Compliant Residual DAGger: Improving Real-World Contact-Rich Manipulation with Human Corrections*. Neural Information Processing Systems (NeurIPS 2025), BEST PAPER AWARD at the Human-to-Robot Workshop (CoRL 2025). [Paper]
5. Haoyu Xiong, **Xiaomeng Xu**, Jimmy Wu, Yifan Hou, Jeannette Bohg, Shuran Song, *Vision in Action: Learning Active Perception from Human Demonstrations*. Conference on Robot Learning (CoRL 2025). [Paper]
6. **Xiaomeng Xu**, Dominik Bauer, Shuran Song, *RoboPanoptes: The All-Seeing Robot with Whole-body Dexterity*. Robotics: Science and Systems (RSS 2025). [Paper]
7. **Xiaomeng Xu**, Huy Ha, Shuran Song, *Dynamics-Guided Diffusion Model for Sensor-less Robot Manipulator Design*. Conference on Robot Learning (CoRL 2024), BEST MACHINE LEARNING PAPER AWARD at the Morphology-Aware Policy and Design Learning Workshop (CoRL 2024). [Paper]
8. **Xiaomeng Xu\***, Yanchao Yang\*, Kaichun Mo, Boxiao Pan, Li Yi, Leonidas Guibas, *JacobiNeRF: NeRF Shaping with Mutual Information Gradients*. Conference on Computer Vision and Pattern Recognition (CVPR 2023). [Paper]
9. Yun Liu\*, **Xiaomeng Xu\***, Weihang Chen, Haocheng Yuan, He Wang, Jing Xu, Rui Chen, Li Yi, *Enhancing Generalizable 6D Pose Tracking of an In-Hand Object with Tactile Sensing*. Robotics

and Automation Letters (RAL 2023), IEEE International Conference on Robotics and Automation (ICRA 2024). [\[Paper\]](#)

10. Xueyi Liu, **Xiaomeng Xu**, Anyi Rao, Chuang Gan, Li Yi, *AutoGPart: Intermediate Supervision Search for Generalizable 3D Part Segmentation*. Conference on Computer Vision and Pattern Recognition (CVPR 2022). [\[Paper\]](#)
11. Guanhong Liu, Tianyu Yu, Zhihao Yao, Haiqing Xu, Yunyi Zhang, Xuhai Xu, **Xiaomeng Xu**, Mingyue Gao, Qirui Sun, Tingliang Zhang, Haipeng Mi, *ViviPaint: Creating Dynamic Painting with a Thermochromic Toolkit*. Multimodal Technologies and Interaction (MTI 2022). [\[Paper\]](#)

\* authors with equal contribution

## INTERNSHIP EXPERIENCE

---

|                                                               |                      |
|---------------------------------------------------------------|----------------------|
| <b>Large Behavior Models Team @ Toyota Research Institute</b> | 06/2025-09/2025      |
| <i>Research Intern, Mentor: Eric Cousineau</i>                | <i>Cambridge, MA</i> |

- Scaling whole-body mobile manipulation with egocentric human demonstrations

|                                                        |                     |
|--------------------------------------------------------|---------------------|
| <b>Stanford Geometric Computing Lab</b>                | 06/2022-08/2022     |
| <i>Research Intern, Advisor: Prof. Leonidas Guibas</i> | <i>Stanford, CA</i> |

- [Chinese Undergraduate Visiting Research Program \(UGVR\)](#).
- Shaping a NeRF to encode mutual correlations of a scene via aligning Jacobians.

## AWARDS

---

|                                                                                                                                                                        |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| NeurIPS 2025 Scholar Award (Travel Grant)                                                                                                                              | 10/2025     |
| Best Paper Award at the Human-to-Robot Workshop (Conference on Robot Learning 2025)                                                                                    | 09/2025     |
| <a href="#">Stanford Interdisciplinary Graduate Fellowship</a> (Awarded to outstanding doctoral students engaged in interdisciplinary research at Stanford University) | 05/2025-now |
| Best Machine Learning Paper Award at the Morphology-Aware Policy and Design Learning Workshop (Conference on Robot Learning 2024)                                      | 11/2024     |
| CVPR 2023 Travel Grant                                                                                                                                                 | 06/2023     |
| Outstanding Graduate (Awarded to top 2% graduates at Tsinghua University)                                                                                              | 06/2023     |
| Comprehensive Excellence Award (Scholarship awarded by Tsinghua University, top 5%)                                                                                    | 10/2022     |
| National Scholarship (Highest scholarship awarded by Chinese Government, top 0.1%)                                                                                     | 10/2021     |
| 129 Scholarship (Highest scholarship for sophomores at Tsinghua University, top 1%)                                                                                    | 10/2020     |

## INVITED TALKS

---

|                                                                                                                                                        |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| From Human Demonstrations to Human Corrections @ <a href="#">RCHI Lab</a> , CMU                                                                        | 04/2026 |
| HoMMI: Learning Whole-Body Mobile Manipulation from Human Demonstrations @ <a href="#">LeCAR Lab</a> , CMU                                             | 04/2026 |
| Learning Whole-Body Manipulation via Hardware-Policy Co-design @ SJTU                                                                                  | 03/2026 |
| Embodiment-Aware Robot Learning: Co-designing Robot Morphology, Sensing, and Control @ <a href="#">The Helping Hands Lab</a> , Northeastern University | 08/2025 |

## SERVICES

---

Teaching assistant of EE/CS227: Robot Perception (Fall 2024-2025)

Co-organizer of RSS 2025 1st Workshop on Robot Hardware-Aware Intelligence, ICRA 2026 Workshop on Reinforcement Learning in the Era of Imitation Learning, RSS 2026 Diffusion for Robot Learning Workshop

Reviewer of RSS, CoRL, T-RO, RA-L, ICRA, IROS, NeurIPS, ICLR, ICML, CVPR, AAAI

## TECHNICAL SKILLS

---

|                           |                                                                       |
|---------------------------|-----------------------------------------------------------------------|
| <b>Computer Languages</b> | Python, C/C++, MATLAB, Verilog/VHDL                                   |
| <b>Software</b>           | SolidWorks, AutoCAD, Rhino, Qt Creator, Multisim, Quartus             |
| <b>Hardware</b>           | 3D Printing, Microcontroller, mechanical design, soldering, woodcraft |
| <b>Tools</b>              | PyTorch, ROS, MuJoCo, IsaacGym                                        |